

PKCERT AI & Software Development Internship

Task 04 – NumPy & Pandas for Data Analysis

Objective:

Develop practical skills in NumPy and Pandas for numerical computing, data manipulation, and exploratory data analysis.

Total Marks: 100

Part A – NumPy Fundamentals (40 Marks)

1. Create one-dimensional and multi-dimensional NumPy arrays using different methods.
2. Perform array indexing, slicing, reshaping, and mathematical operations.
3. Demonstrate broadcasting using practical examples and explain its advantages.
4. Implement vectorized operations and compare them with traditional loops.
5. Perform linear algebra operations including dot product, matrix multiplication, transpose, and inverse (where applicable).

Part B – Pandas Fundamentals (30 Marks)

1. Create and manipulate Pandas Series and DataFrames.
2. Perform indexing, filtering, sorting, and selection operations.
3. Apply groupby operations to summarize datasets.
4. Merge and join multiple DataFrames and explain the results.

Part C – Data Analysis Mini Project (30 Marks)

Select a public CSV dataset (such as the Titanic dataset or any other public dataset). Develop a Python application that reads the dataset, cleans missing values, generates summary statistics, and performs basic exploratory data analysis using Pandas. Prepare a README describing the dataset, data cleaning process, analysis performed, and key findings. Upload the complete project to your GitHub repository.

Evaluation Criteria

NumPy Fundamentals	40
Pandas Fundamentals	30
Mini Project	30

Total

100